

This Month's DVD Contents

This month's DVD contains some interesting electronics design automation (EDA) tools for design and simulation, programming tools and informative videos complemented with some useful resources and fun applications

ABHISHEK A. MUTHA

CadSoft EAGLE (version 7.3.0)

EAGLE PCB design software is a popular tool of choice for many design engineers and academia across the world. It comprises features like a schematic capture editor, PCB layout editor and an auto router, to name a few. Make sure you select *Run as Freeware* option to access EAGLE Light Edition for free and non-commercial or evaluation usage.

KEMET SPICE (version 4.0.3)

This software is aimed at supporting design engineers in analysing the performance of capacitors over direct current (DC) bias conditions, ripple, temperature and frequency. The different kinds of capacitors include polymer, aluminium polymer ceramic (MLCC) and tantalum. Although it is a stand-alone Windows based program, it is able to create files that can be imported into some of the commercial versions of SPICE.

Fritzing (version 0.9.2b)

In the spirit of processing and Arduino, Fritzing is an open source software tool that makes electronics accessible as a creative material for anyone. It allows you to document and share your prototypes with others, impart electronics lessons in a classroom and layout as well as manufacture professional PCBs. This version brings you many latest new popular parts such as Raspberry Pi 2, RaspIO Duino, STM32 Nucleo and many more.

Rapid SCADA (version 4.4)

Rapid SCADA is a free, open source and full-featured supervisory control and data acquisition (SCADA) software. Using Rapid SCADA you can

Now, run all software on EFY+ DVD irrespective of OS requirements

With Oracle VirtualBox 5.0 bundled with this month's DVD, you can now run any software tool and legacy applications in a virtual environment. VirtualBox runs on Windows, Linux, Macintosh and Solaris hosts, and supports a large number of guest operating systems including, but not limited to, Windows (NT 4.0, 2000, XP Server 2003, Vista, Windows 7 and Windows 8), DOS/Windows 3.x, Linux (2.4, 2.6 and 3.x), Solaris and OpenSolaris, OS/2 and OpenBSD.

Some popular resources

- ClamWin (version 0.98.7)
- Expert PDF Reader
- VirtualBox (version 5.0)
- WinDirStat (version 1.1.2)
- Wireshark (version 1.12.6)

create automated systems such as industrial automation systems, home automation systems, energy accounting systems and any other systems that contain controllers, sensors and relays. The latest version contains all primary Rapid SCADA applications, a set of basic drivers and the configuration database with ready examples.

Qucs (version 0.0.18)

Quite Universal Circuit Simulator (Qucs) is an integrated circuit simulator, which allows you to set up a circuit with a graphical user interface (GUI) and simulate the large-signal, small-signal and noise behaviour of the circuit. Post simulation, you can view the results on a presentation page or window. The GUI is based on Qt by Digia. The software aims to support all kinds of circuit simulation types such as S-parameter, noise analysis, alternating current (AC), harmonic balance analysis and DC, to name a few.

IT++ (version 4.3.1)

IT++ is a C++ library of mathematical, signal processing and communication classes and functions. Its main use is in simulation of communication systems and for

Other tools

- Tiny Multi Bootloader+ (v0.10.0). Serial bootloader for Microchip, Atmel, NXP, TI, 8051 microcontrollers (MCUs)
- Magic VLSI layout tool (version 8.0.207), SVEditor (beta version), OpenCV (version 3.0), Universal JTAG library, server and tools (UrJTAG version 0.10) and PPMScope (MCU based PC oscilloscope)

performing research in the area of communications. In order to use all functionality provided by IT++ library, it is recommended that you have some external libraries compiled and installed on your computer. This is why we have also provided the basic set of those external libraries, namely, BLAS, LAPACK and FFTW (version 3.3.4) with IT++.

Programming Without Coding Technology (version 1.9)

Rather than creating your application in a few steps, Programming Without Coding Technology (PWCT) is a general-purpose visual programming tool designed for novice and expert programmers. While a novice can use it to learn programming concepts such as data structure, control structure or programming paradigm, an expert can use it to develop large and/or complex software. With support for languages like Harbour, Supernova, C, Python and C#.NET, you can see/edit the generated source code. ●

The author is a senior technical correspondent at EFY